Last-Mile Route Optimization Using K-Means Clustering and Geospatial Analytics

Introduction:

Last-mile delivery refers to the final stage of the supply chain where goods are transported from a distribution center or fulfillment hub to the customer's doorstep. Although it is often the shortest physical segment of the delivery journey, it has become the most expensive, inefficient, and operationally difficult part of modern logistics. The rapid growth of e-commerce has dramatically increased delivery volumes, forcing logistics companies to manage millions of individual residential deliveries every day while customers simultaneously expect shipping to be faster, cheaper, and more reliable than ever before. Industry research shows that last-mile delivery now accounts for approximately 53% of total shipping costs, making it the single largest financial burden within the delivery process.

The core problem this project addresses is the inefficiency of modern delivery routing systems. Many delivery operations still rely heavily on static scheduling systems or basic GPS navigation that fail to account for traffic patterns, delivery density, geographic clustering, and real-time optimization opportunities. As a result, drivers frequently travel unnecessary miles, spend excessive time in traffic, revisit overlapping service zones, and complete routes that are far from optimal. These inefficiencies directly increase fuel consumption, labor costs, delivery times, and operational expenses. With the average delivery costing companies more to complete than customers are willing to pay, inefficient routing creates a growing financial strain that threatens the sustainability of large-scale last-mile operations.

Beyond financial costs, inefficient last-mile delivery also creates significant environmental and social challenges. Every unnecessary mile traveled by a delivery vehicle contributes additional carbon emissions, traffic congestion, and fuel consumption at a time when governments and consumers are placing increasing pressure on companies to adopt more sustainable business practices. In addition, routing inefficiencies often create unequal service quality across neighborhoods, where lower-density or geographically inconvenient areas may experience slower deliveries, wider delivery windows, and reduced service reliability. As e-commerce volumes continue to rise globally, these problems are becoming increasingly difficult for logistics providers to ignore.

This project aims to address these challenges by using data-driven route optimization techniques to improve delivery efficiency, reduce operational costs, lower emissions, and identify underserved delivery areas. Using real Amazon delivery data, the project applies K-Means clustering, OR-Tools route optimization, geospatial analysis, and emissions modeling to design more efficient delivery routes and evaluate their operational impact. The goal is not simply to analyze delivery patterns academically, but to develop actionable insights and optimization strategies that logistics companies could realistically implement to improve the performance, sustainability, and fairness of last-mile delivery systems.

Data Overview and Data Cleaning:

The Amazon Last Mile Routing Research Challenge dataset, released jointly by Amazon and the Massachusetts Institute of Technology (MIT), contains real delivery route information collected across five major United States metropolitan areas during 2018. The raw dataset included millions of GPS coordinates, route records, and stop-level delivery details that required extensive preprocessing before meaningful analysis could be performed.

To ensure analytical accuracy and eliminate invalid operational data, a seven-step cleaning pipeline was applied. The final processed dataset retained 898,415 delivery stops across 6,112 complete delivery routes, making it one of the largest real-world last-mile logistics datasets publicly available.

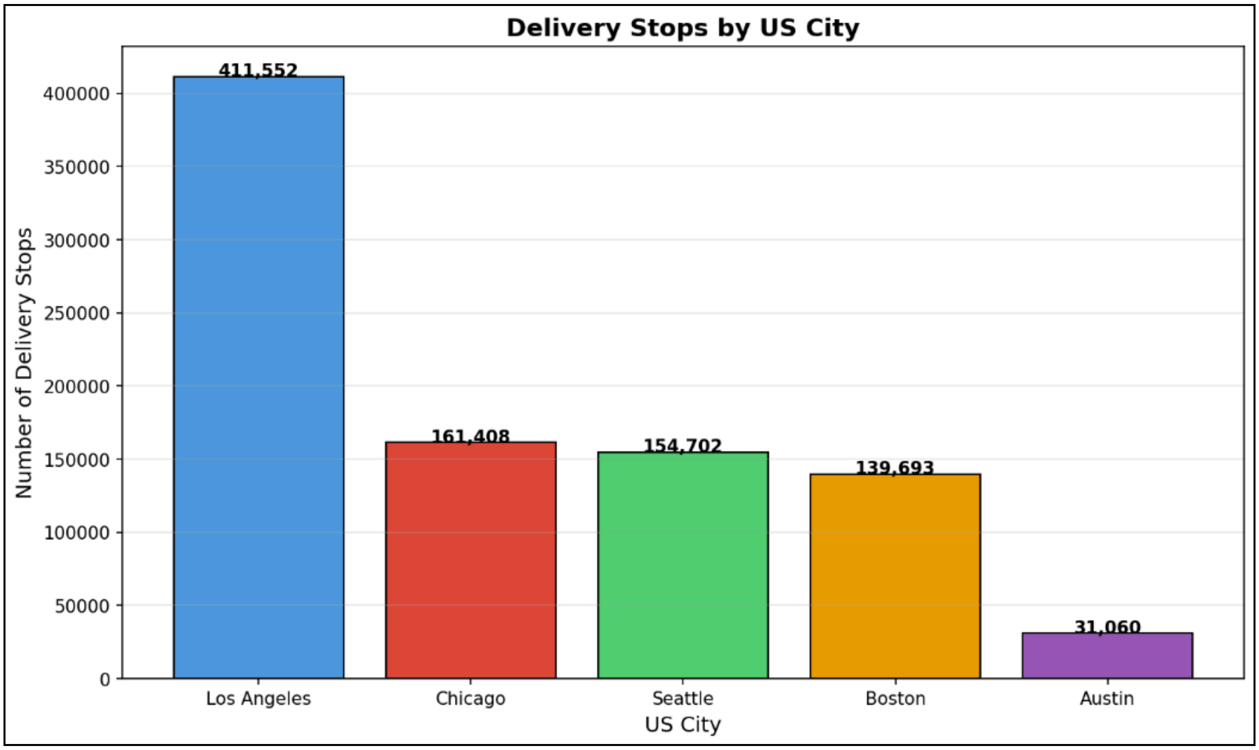
The cleaning process removed depot-only records, duplicate delivery stops, missing GPS coordinates, invalid coordinate values such as (0,0), routes outside continental US boundaries, unidentified city codes, routes with fewer than three stops, and extreme outlier distances likely caused by GPS or logging errors. These preprocessing steps ensured the final dataset accurately reflected real Amazon delivery operations while removing data anomalies that could distort clustering and route optimization results.

Geographic Distribution of Delivery Activity:

The cleaned dataset includes delivery activity from five operationally distinct metropolitan areas:

- Los Angeles, CA
- Chicago, IL
- Seattle, WA
- Boston, MA
- Austin, TX

Delivery activity was not evenly distributed across cities. Los Angeles dominated the dataset with nearly half of all delivery stops, reflecting its role as a major logistics and fulfillment hub for Amazon operations. Austin represented the smallest share of the dataset, suggesting either a smaller operational footprint or a less mature delivery network during the period of data collection.



The delivery stop distribution chart clearly demonstrates the imbalance in operational scale between cities. Los Angeles recorded approximately **411,552 delivery stops**, representing nearly **46% of the entire dataset**. Chicago and Seattle followed with roughly 161,000 and 154,000 stops respectively, while Boston contributed approximately 140,000 stops. Austin contained only about 31,000 delivery stops.

This imbalance is important because larger delivery volumes typically create opportunities for denser route clustering and improved operational efficiency. High-density delivery environments allow drivers to complete more stops within shorter geographic distances, reducing travel time between deliveries. In contrast, lower-density markets often require vehicles to travel farther between stops, increasing route inefficiency and fuel consumption.

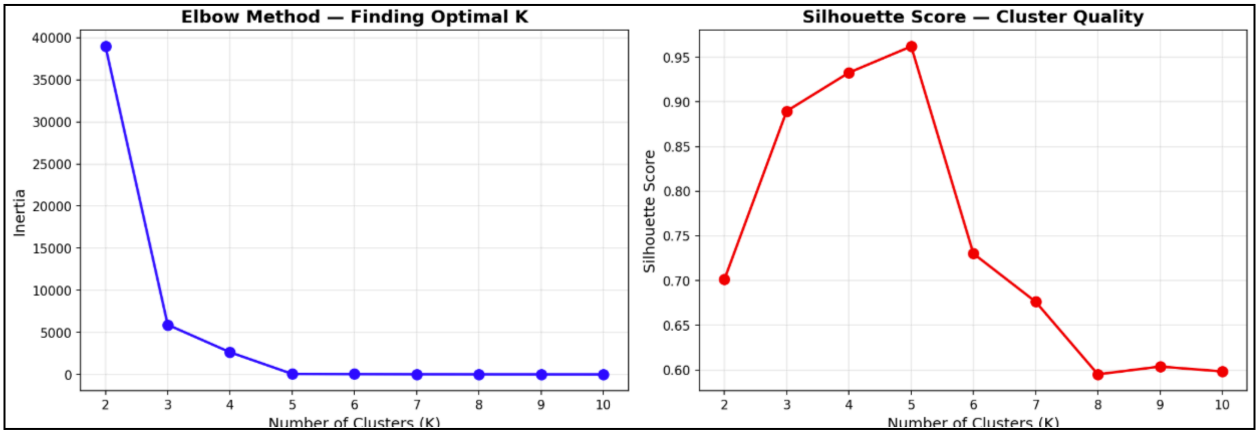
The operational scale differences shown in the figure help explain later findings related to route distance and emissions variation between cities.

K-Means Clustering & Delivery Zone Identification

To identify natural delivery territories within the dataset, K-Means clustering was applied to a random sample of 50,000 delivery coordinates. The objective was to determine whether geographic delivery zones emerge naturally from the stop data without requiring manually defined territories.

Two standard clustering evaluation techniques were used:

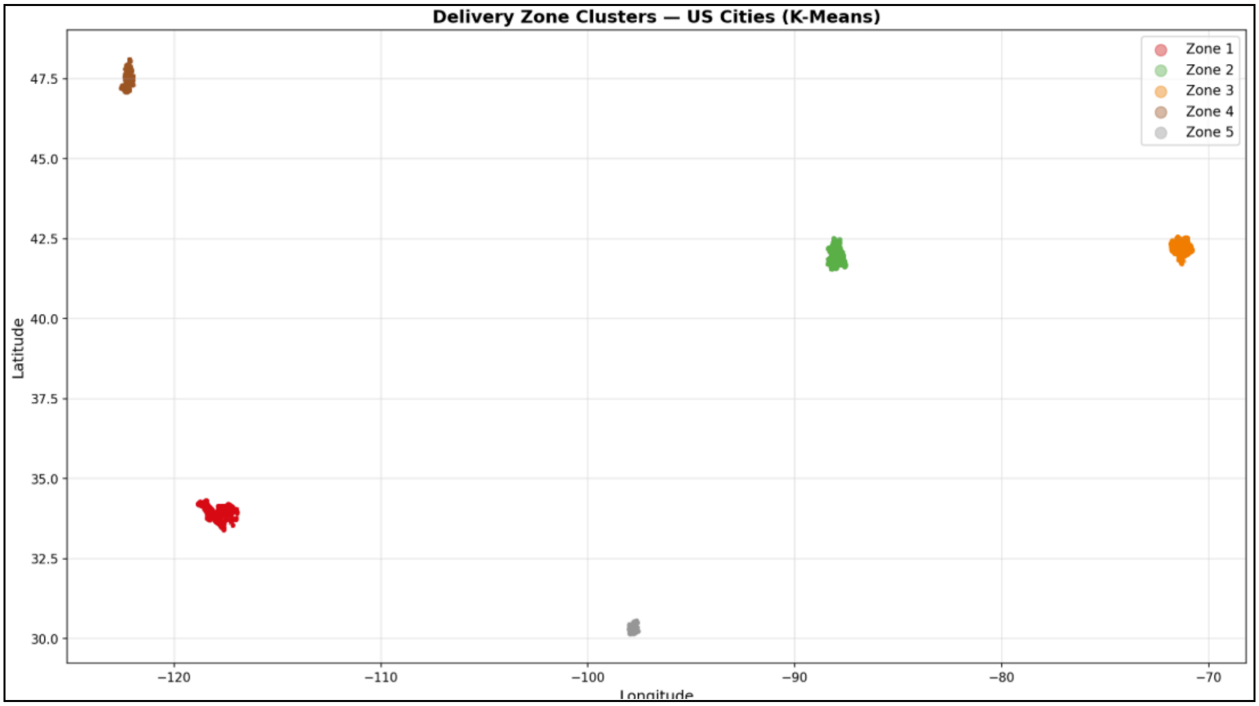
1. **Elbow Method**
2. **Silhouette Score Analysis**



The Elbow Method graph shows a steep reduction in inertia values between K=2 and K=5 clusters, after which improvements flatten significantly. This indicates diminishing optimization benefits beyond five clusters.

The Silhouette Score graph provides even stronger validation. The silhouette score peaks at approximately **0.96 for K=5**, which is exceptionally high for real-world geospatial clustering problems. Scores decline sharply beyond K=5, indicating worsening cluster separation and overlap.

Together, these results mathematically confirm that the optimal number of clusters is five, matching the five metropolitan regions included in the dataset.



The K-Means clustering visualization demonstrates that the algorithm independently separated delivery stops into five perfectly distinct geographic zones corresponding to the five cities in the dataset. There is essentially zero overlap between clusters, indicating extremely strong geographic separation.

This finding is operationally significant because it validates the use of clustering algorithms for automated territory design in large-scale delivery systems. Rather than relying on manually drawn administrative boundaries, logistics companies can use coordinate-driven clustering techniques to identify natural delivery zones that minimize cross-territory travel.

The clustering results also demonstrate that geographic delivery density strongly influences operational efficiency. Cities with tighter geographic concentration naturally form compact delivery territories that reduce unnecessary travel distance.

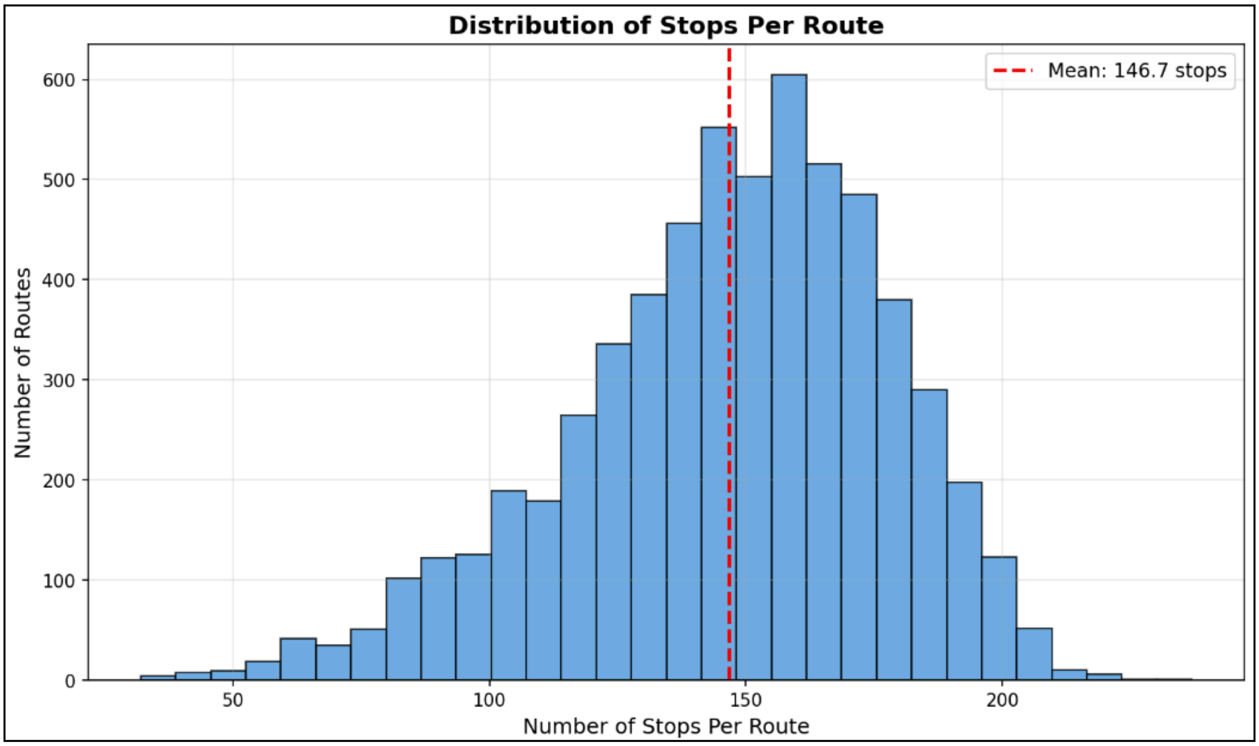
Route Structure and Distribution:

After cleaning and route validation, the dataset contained **6,112 complete delivery routes**. Analysis of stop counts revealed a highly standardized operational structure across Amazon's delivery network.

Key route statistics include:

- Average stops per route: **146.7**

- Minimum stops: **32**
- Maximum stops: **237**
- Standard deviation: **31 stops**

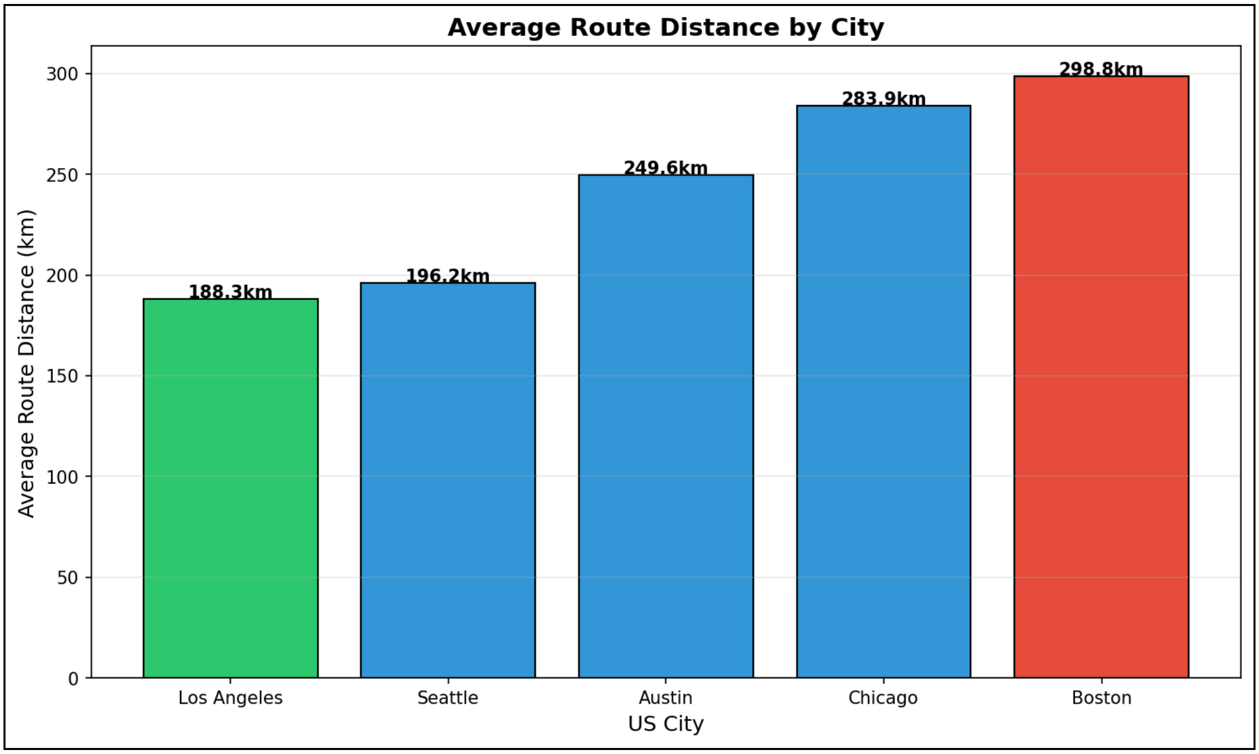


The stop distribution histogram follows a near-normal bell curve centered around the mean of 146.7 stops. Most routes fall within a relatively narrow range around this average, indicating a highly standardized delivery operation.

This consistency is operationally important because it removes delivery volume as the primary explanation for route inefficiency differences between cities. Since most drivers complete a similar number of stops regardless of city, variations in route distance and emissions are more likely caused by geographic layout, routing quality, and stop sequencing efficiency rather than workload differences.

The tight clustering around the mean also means optimization improvements can scale predictably across the network. For example, reducing average route distance by 10% would likely produce consistent fuel and emissions savings across nearly every route in the fleet.

Route Distance Inefficiency by City



The route distance comparison reveals dramatic operational inefficiencies between cities.

Los Angeles recorded the shortest average route distance at approximately **188.3 km**, while Boston recorded the highest at nearly **298.8 km**. Chicago also showed high route distances at approximately **283.9 km**, while Austin averaged about **249.6 km**.

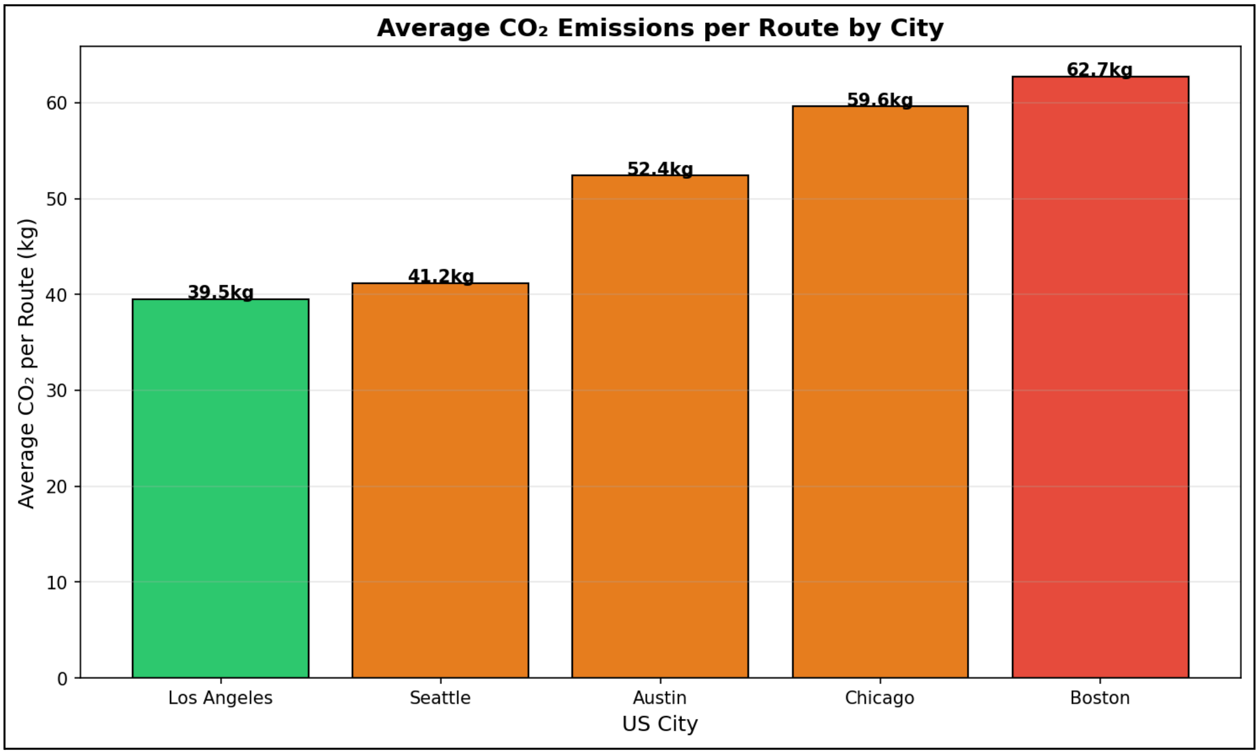
The most significant result is the difference between Boston and Los Angeles. Drivers in Boston travel roughly **110 km more per route** while completing approximately the same number of delivery stops.

This finding strongly suggests that route inefficiency is driven primarily by geography and urban structure rather than delivery volume. Boston's historic street design, irregular road layouts, water barriers, and dense urban constraints force drivers into less direct travel patterns. In contrast, Los Angeles benefits from denser stop clustering and more structured road networks despite having significantly larger delivery volumes.

The implication is that route optimization strategies should prioritize cities where geographic inefficiencies create excessive travel distances.

<u>CO2 emission Analysis:</u>

To evaluate environmental impact, average CO_2 emissions per route were estimated using EPA SmartWay delivery vehicle emission factors of approximately **0.21 kg CO_2 per kilometer traveled**.



The emissions analysis mirrors the route distance findings almost exactly because travel distance is the primary driver of delivery vehicle emissions.

Los Angeles generated the lowest emissions at approximately **39.5 kg CO_2 per route**, while Boston generated the highest at approximately **62.7 kg CO_2 per route**. Chicago followed closely at approximately **59.6 kg CO_2 per route**.

Boston therefore produces approximately **23.2 kg more CO_2 per route** than Los Angeles while delivering a similar number of packages. This represents a **59% increase in emissions efficiency loss** caused almost entirely by longer route distances.

This result is especially important because it demonstrates that major emissions reductions can be achieved without purchasing new electric vehicles or expanding infrastructure. Simply improving routing efficiency and reducing unnecessary travel distance can substantially lower environmental impact.

If Boston routes were optimized to match Los Angeles route efficiency, the city could eliminate tens of thousands of kilograms of CO_2 emissions annually across its delivery fleet.

Overall Findings:

The analysis revealed several major findings with direct implications for delivery efficiency, operational cost reduction, and sustainability planning within last-mile logistics systems.

Geographic clustering is highly effective for delivery zone design

K-Means clustering successfully identified the five metropolitan delivery regions with a silhouette score of 0.96, indicating extremely strong cluster separation. The results demonstrate that natural delivery territories emerge directly from geographic stop patterns, validating clustering as a reliable method for automated territory planning.

Route inefficiency varies substantially between cities

Despite maintaining nearly identical average stop counts per route, cities showed major differences in average travel distance. Boston routes were approximately 59% longer than Los Angeles routes, suggesting that urban structure, road layout, and routing strategy strongly influence delivery efficiency.

CO_2 emissions are directly tied to routing efficiency

Cities with longer delivery routes consistently produced higher emissions per route. Boston generated approximately 23 kg more CO_2 per route than Los Angeles while completing a similar delivery workload, demonstrating that inefficient routing significantly increases environmental impact.

Amazon’s route operations are highly standardized

The stop-per-route distribution followed a tightly centered bell curve around an average of 146.7 stops per route. This operational consistency means optimization improvements can scale predictably across the delivery network, strengthening the business case for route optimization technologies.

Delivery density improves operational efficiency

Cities with denser delivery clustering generally achieved shorter route distances and lower emissions. Los Angeles, despite having the highest delivery volume in the dataset, maintained the shortest average route distance, showing that high-density delivery environments can improve routing efficiency when properly optimized.

Recommendations:

Based on the analysis findings, several operational and strategic recommendations emerge for improving last-mile delivery efficiency and sustainability.

1. Prioritize Boston for Route Optimization

Boston represents the largest opportunity for operational improvement due to its extremely high average route distance and CO_2 emissions. Redesigning delivery territories, improving stop sequencing, and optimizing depot coverage could significantly reduce unnecessary travel distance and fuel consumption.

2. Use Los Angeles as an Efficiency Benchmark

Los Angeles demonstrates that high-volume delivery operations can still maintain relatively efficient route distances when stop density and route structure are optimized effectively. Other metropolitan areas should benchmark routing performance against Los Angeles operational metrics.

3. Expand Geographic Clustering for Territory Design

The clustering analysis showed that natural delivery zones emerge directly from delivery stop coordinates. Logistics companies should use clustering algorithms to design territories based on real delivery density patterns rather than static administrative boundaries.

4. Improve Depot Placement in Austin

Austin’s relatively low delivery density combined with comparatively high route distances suggests that current depot placement may not fully support efficient delivery coverage. Expanding distribution infrastructure or repositioning fulfillment hubs could reduce travel distances and improve scalability as demand grows.

Conclusion:

Last-mile delivery has become one of the most expensive and operationally difficult components of modern supply chain management due to the rapid growth of e-commerce and rising customer expectations for fast, low-cost delivery. Using real Amazon delivery data, this project demonstrated that routing efficiency is influenced more by geographic structure, delivery density, and route design than by delivery volume alone. Through clustering analysis, route distance evaluation, and emissions modeling, the results showed that cities with inefficient routing structures experience significantly longer travel distances, higher operational costs, and greater environmental impact despite maintaining similar delivery workloads.

The analysis revealed that Boston delivery routes were approximately 59% longer than Los Angeles routes while servicing nearly the same number of stops, leading to substantially higher fuel consumption and CO_2 emissions. At the same time, the project showed that many of these inefficiencies can be reduced through data-driven optimization techniques such as geographic clustering, smarter territory design, and improved route sequencing. These findings highlight the importance of advanced routing systems in improving both the operational efficiency and sustainability of last-mile logistics networks.

As e-commerce demand continues to grow globally, logistics companies will increasingly rely on data analytics, machine learning, and real-time optimization technologies to reduce costs while meeting sustainability goals and customer expectations. This project demonstrates that significant improvements in delivery performance and environmental impact are achievable through more intelligent use of routing data and geographic analytics, making route optimization a critical component of the future of last-mile delivery systems.